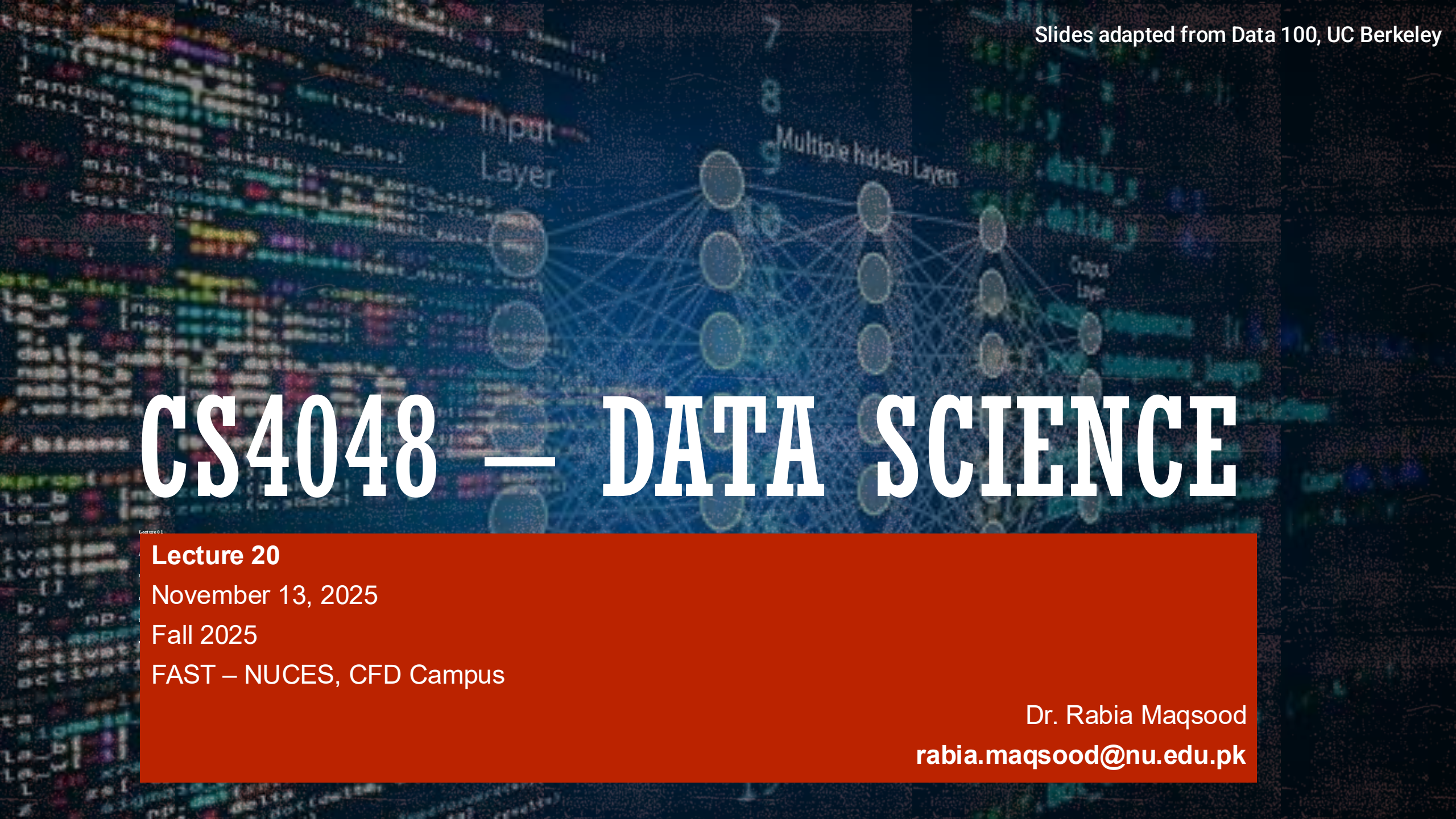


CS4048 — DATA SCIENCE



Lecture 20

November 13, 2025

Fall 2025

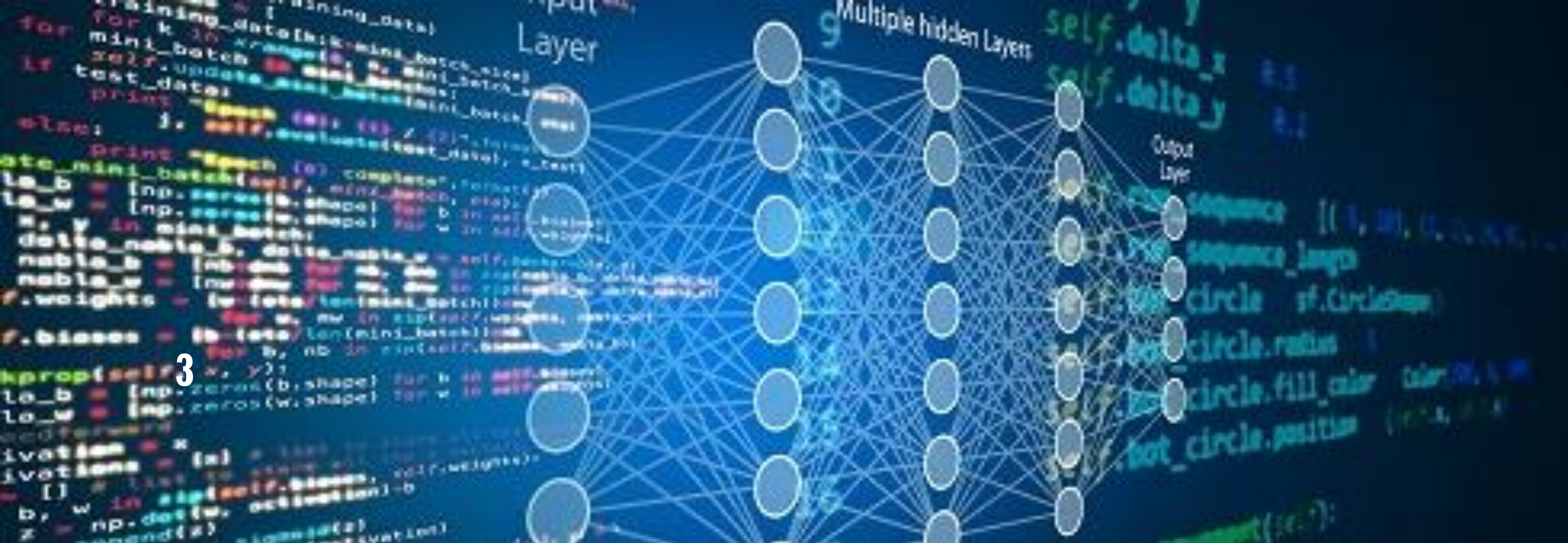
FAST – NUCES, CFD Campus

Dr. Rabia Maqsood

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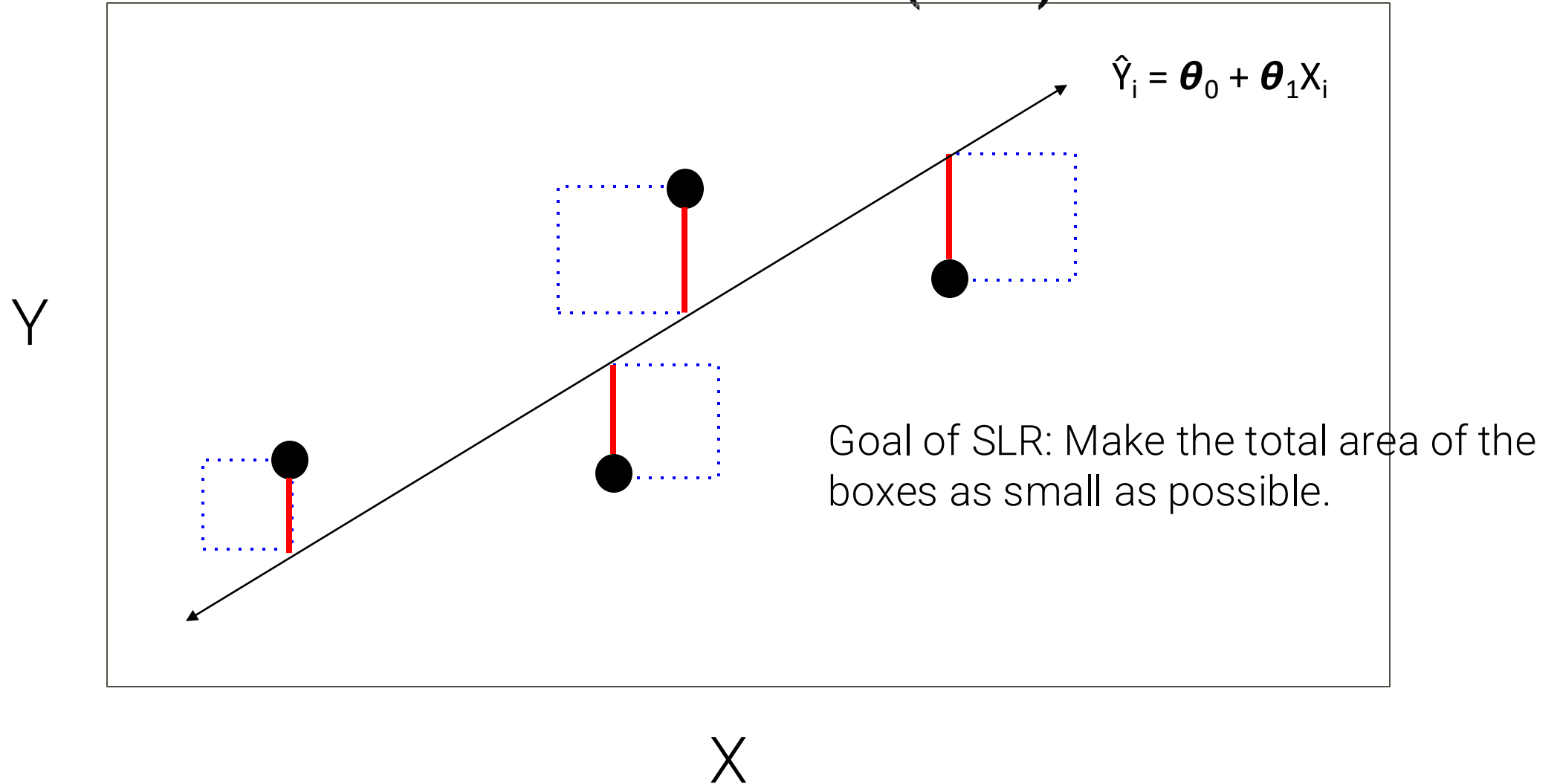
TODAY'S TOPICS

- Simple Linear Regression
- Constant model (baseline): continued
 - using MSE & MAE loss functions
- Transformations of linear models

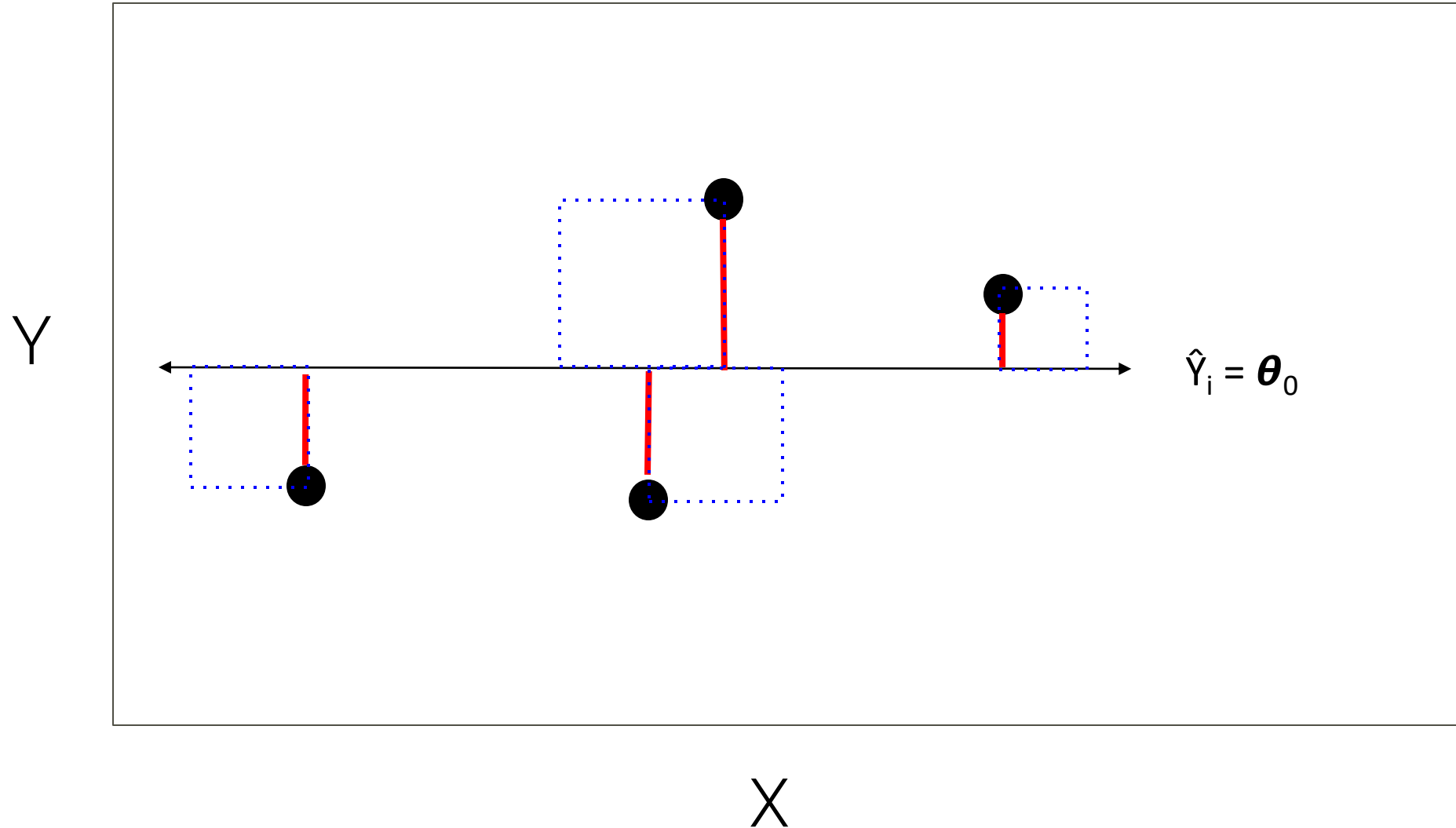


CONSTANT MODEL + MSE

VISUALIZING THE SUM OF SQUARED ERROR FOR SIMPLE LINEAR REGRESSION (SLR)



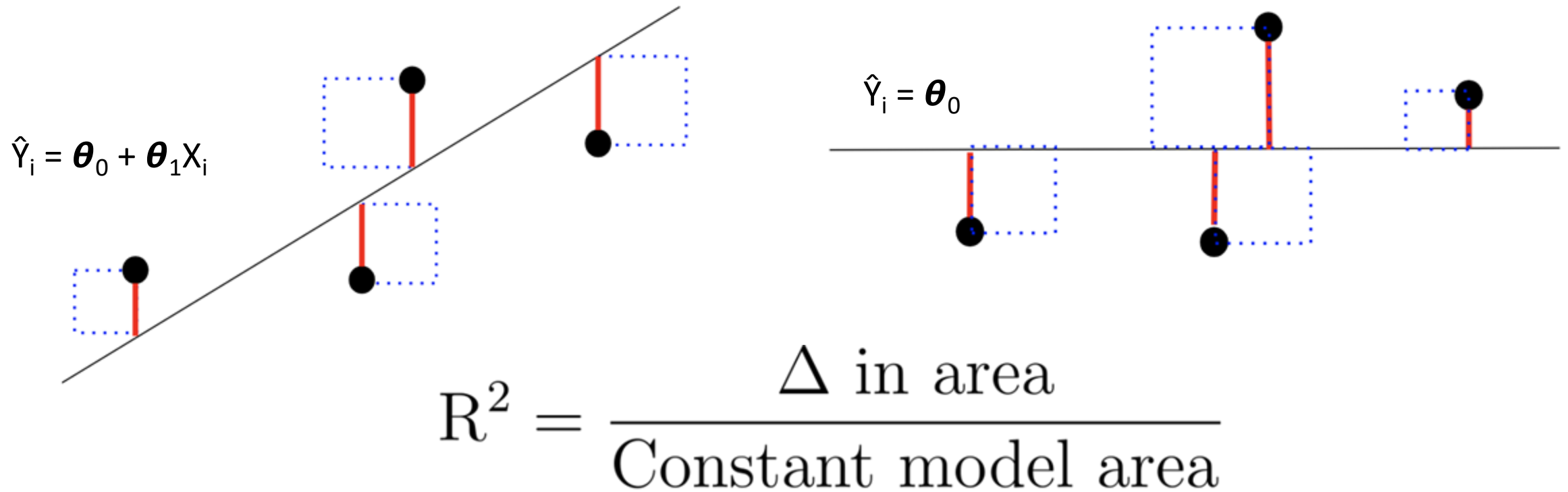
VISUALIZING THE SUM OF SQUARED ERROR FOR A CONSTANT MODEL



Also known as an intercept-only model.



R² MEASURES QUALITY OF LINEAR MODEL FIT RELATIVE TO CONSTANT MODEL



THE MODELING PROCESS: USING A DIFFERENT MODEL

1. Choose a model

~~SLR model~~
 ~~$\hat{y} = \theta_0 + \theta_1 x$~~

Constant Model $\hat{y} = \theta_0$

2. Choose a loss function

L2 Loss

Mean Squared Error
(MSE)

3. Fit the model

Minimize
average loss
with calculus

4. Evaluate model performance

THE CONSTANT MODEL

You work at a local boba shop and want to predict the daily number of drinks sold.

Here's your data from 5 randomly selected previous days:

$\{20, 29, 22, 21, 33\}$

You have other info (e.g., the weather and day of week), but you *ignore everything except the historical # of drinks sold*.

Your model always predicts the same number.

This is a **constant model**.



THE CONSTANT MODEL

The **constant model** always predicts the same number—i.e., a constant.

It ignores any relationships between variables:

- For instance, boba tea sales likely depend on the time of year, the weather, how the customers feel, whether school is in session, etc.
- Ignoring these factors is a **simplifying assumption**.

Like SLR, the constant model is a parametric, statistical model:

$$\hat{y} = \theta_0$$

THE CONSTANT MODEL

Our parameter θ_0 is 1-dimensional. $\theta_0 \in \mathbb{R}$

Unlike SLR, no input needed to generate a prediction; we always predict $\hat{y} = \theta_0$

Like before, we can still determine the best θ_0 that minimizes **average loss** on our data.

$$\hat{y} = \theta_0$$

THE MODELING PROCESS: USING A DIFFERENT MODEL



1. Choose a model

SLR model
 ~~$\hat{y} = \theta_0 + \theta_1 x$~~

Constant Model $\hat{y} = \theta_0$

2. Choose a loss function

L2 Loss

Mean Squared Error (MSE)

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

"Average loss"
"Objective function"
"Mean squared error"

3. Fit the model

Minimize
average loss
with calculus

4. Evaluate model performance

Why is there a hat above R? We are *estimating* the average loss with a sample. Estimated values often have a hat. Technically, there is also a "true" loss for the entire population from which our sample is drawn. We would write the expression for this "true" loss without a hat. Don't worry about this for Data 100!

THE MODELING PROCESS: USING A DIFFERENT MODEL

1. Choose a model



~~SLR model~~
 ~~$\hat{y} = \theta_0 + \theta_1 x$~~

Constant Model $\hat{y} = \theta_0$

2. Choose a loss function



L2 Loss
Mean Squared Error
(MSE)

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

3. Fit the model

Minimize
average loss
with calculus

How does this step change?

4. Evaluate model performance

Why is there a hat above R? We are *estimating* the average loss with a sample. Estimated values often have a hat. Technically, there is also a "true" loss for the entire population from which our sample is drawn. We would write the expression for this "true" loss without a hat. Don't worry about this for Data 100!

FIT THE MODEL: REWRITE MSE FOR THE CONSTANT MODEL

Recall that Mean Squared Error (MSE) is average squared loss (L2 loss) over the data $\mathcal{D} = \{y_1, y_2, \dots, y_n\}$:

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \underbrace{(y_i - \hat{y}_i)^2}_{\text{L2 loss on a single datapoint}}$$

L2 loss on a
single
datapoint

Given the **constant model** $\hat{y} = \theta_0$:

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2$$

We **fit the model** by finding the optimal $\hat{\theta}_0$ that minimizes the MSE.

FIT THE MODEL: WHERE WE ARE HEADED

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2$$

With calculus, we can show that the average loss (i.e., our objective function) is minimized by

$$\hat{\theta}_0 = \text{mean}(y) = \bar{y}$$

Given a dataset, we can also find the minimizing θ_0 with an iterative algorithm.

FIT THE MODEL: CALCULUS FOR THE GENERAL CASE

1. Differentiate with respect to θ_0 :

$$\frac{d}{d\theta_0} \hat{R}(\theta_0) = \frac{d}{d\theta_0} \left(\frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2 \right)$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{d}{d\theta_0} (y_i - \theta_0)^2$$

Derivative of sum is
sum of derivatives

$$= \frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0)(-1)$$

Chain rule

$$= \frac{-2}{n} \sum_{i=1}^n (y_i - \theta_0)$$

Simplify constants

2. Set equal to 0.

$$0 = \frac{-2}{n} \sum_{i=1}^n (y_i - \hat{\theta}_0)$$

3. Solve for $\hat{\theta}_0$.

$$0 = \cancel{\frac{-2}{n}} \sum_{i=1}^n (y_i - \hat{\theta}_0) = \sum_{i=1}^n (y_i - \hat{\theta}_0)$$

$$= \sum_{i=1}^n y_i - \sum_{i=1}^n \hat{\theta}_0$$

Separate sums

$$= \sum_{i=1}^n y_i - n \cdot \hat{\theta}_0$$

c + c + ... + c = nxc

$$n \cdot \hat{\theta}_0 = \sum_{i=1}^n y_i$$

$$\hat{\theta}_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i \right) \implies \boxed{\hat{\theta}_0 = \bar{y}}$$

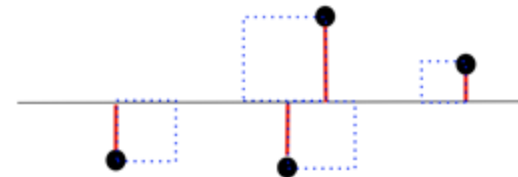
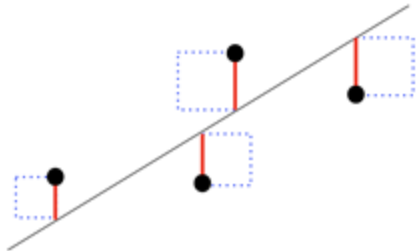
INTERPRETING $\hat{\theta}_0 = \bar{y}$

The **mean** of the outcomes achieves the minimum MSE of the constant model.

Minimum MSE is the **sample variance** of $y \rightarrow R(\hat{\theta}_0) = R(\bar{y}) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 = \sigma_y^2$

R^2 (from earlier) is interpreted as the fraction of **variance in y** "explained" by a linear model.

$$R^2 = \frac{\Delta \text{ in area}}{\text{Constant model area}} = \frac{\text{MSE}_{\text{constant model}} - \text{MSE}_{\text{linear model}}}{\text{MSE}_{\text{constant model}}}$$



Technical note: The area is the sum of squared errors, not the MSE. $\text{Area}/n = \text{MSE}$. Also, the R in the average loss expression is unrelated to R^2 .

Why "explained" is in quotes: There is no causality here!

INTERPRETING $\hat{\theta}_0 = \bar{y}$

$$R(\hat{\theta}_0) = \min_{\theta_0} R(\theta_0) = \sigma_y^2 \quad \text{vs} \quad \hat{\theta}_0 = \operatorname{argmin}_{\theta_0} R(\theta_0) = \bar{y}$$

The **minimum** MSE

The **arg**ument that **min**imizes MSE

In modeling, we care less about **minimum loss** $R(\hat{\theta}_0)$ and more about the **minimizer** of loss.

THE MODELING PROCESS: USING A DIFFERENT MODEL

1. Choose a model



Constant Model

Constant Model $\hat{y} = \theta_0$

2. Choose a loss function



L2 Loss

Mean Squared Error (MSE)

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2$$

3. Fit the model



Minimize average loss with calculus

$$\hat{\theta}_0 = \text{mean}(y) = \bar{y}$$

4. Evaluate model performance

[DATA] COMPARING TWO DIFFERENT MODELS, BOTH FIT WITH MSE

Modeling the relationship of dugong age (y) and length (x).



A Dugong [[image source](#)]



Not a Dugong, a Dewgong [[image source](#)]

Constant Model

$$\hat{y} = \theta_0$$

Data: Sample of ages.

$$\mathcal{D} = \{y_1, y_2, \dots, y_n\}$$

Simple Linear Regression

$$\hat{y} = \theta_0 + \theta_1 x$$

Data: Sample of (length, age)s.

$$\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

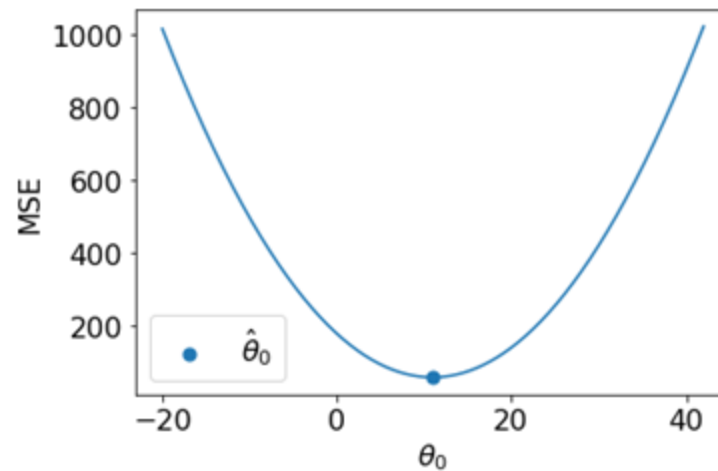
[LOSS] COMPARING TWO DIFFERENT MODELS, BOTH FIT WITH MSE

Constant Model

$$\hat{y} = \theta_0$$

is **1-D**.

$\hat{\theta}_0$ Loss surface is **2-D**.



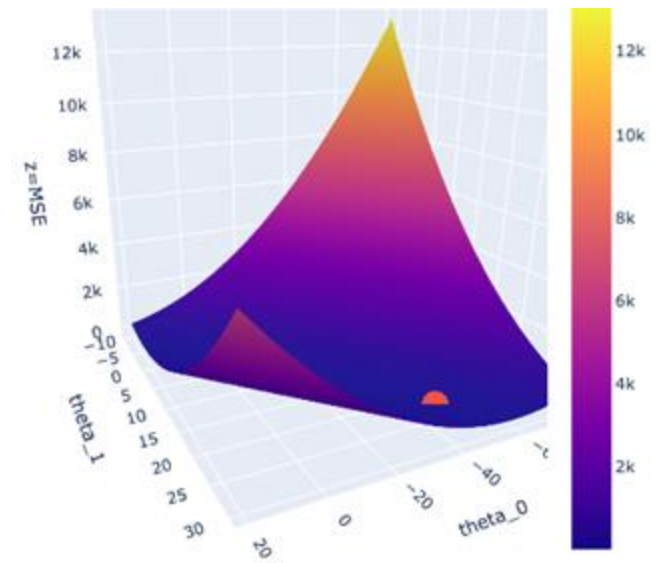
$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2$$

Simple Linear Regression

$$\hat{y} = \theta_0 + \theta_1 x$$

$\hat{\theta} = [\hat{\theta}_0, \hat{\theta}_1]$ is **2-D**.

Loss surface is **3-D**.



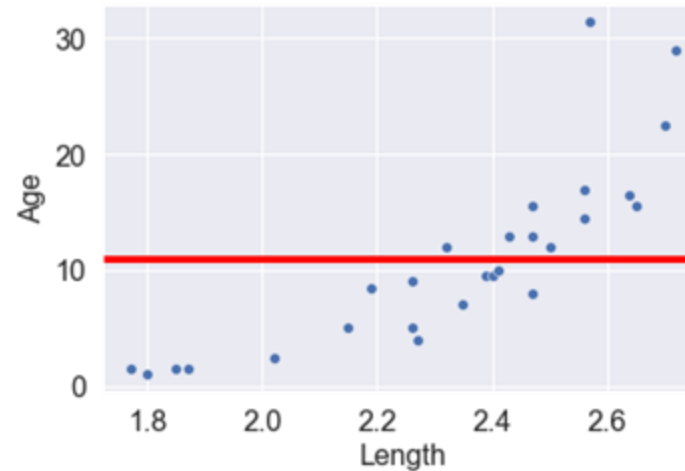
$$\hat{R}(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x))^2$$

[FIT] COMPARING TWO DIFFERENT MODELS, BOTH FIT WITH MSE

Constant Model

$$\hat{y} = \theta_0$$

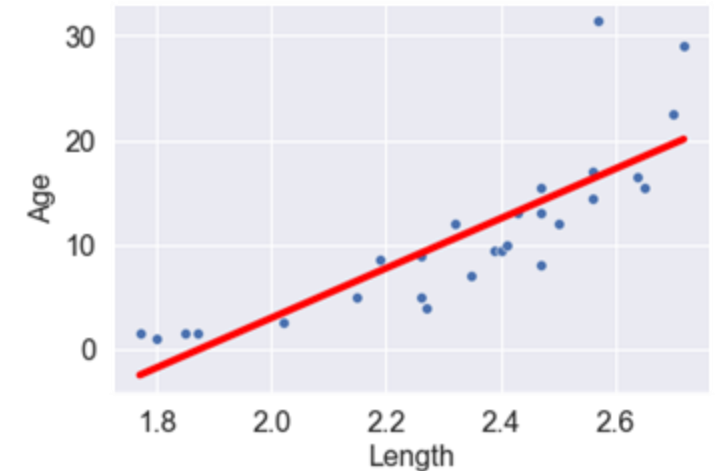
Root MSE (RMSE): **7.72**



Simple Linear Regression

$$\hat{y} = \theta_0 + \theta_1 x$$

Root MSE (RMSE): **4.31**



Interpretation of RMSE:

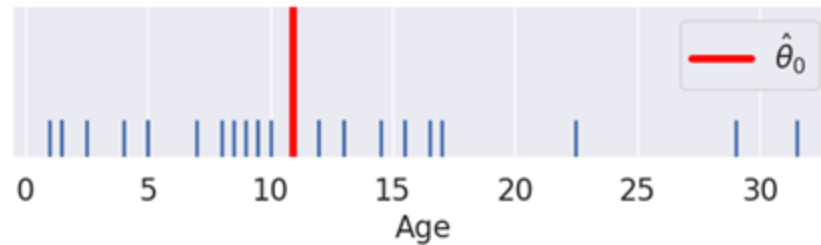
- Constant model RMSE is **higher** than linear model RMSE
- Constant model has a **poorer fit** than linear model, using RMSE as our metric.

[FIT] COMPARING TWO DIFFERENT MODELS, BOTH FIT WITH MSE

Constant Model

$$\hat{y} = \theta_0$$

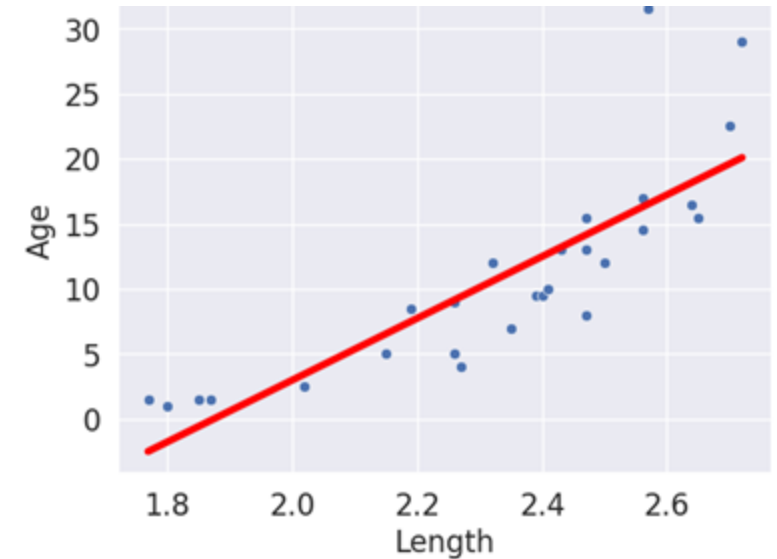
Predictions on a **rug plot**.



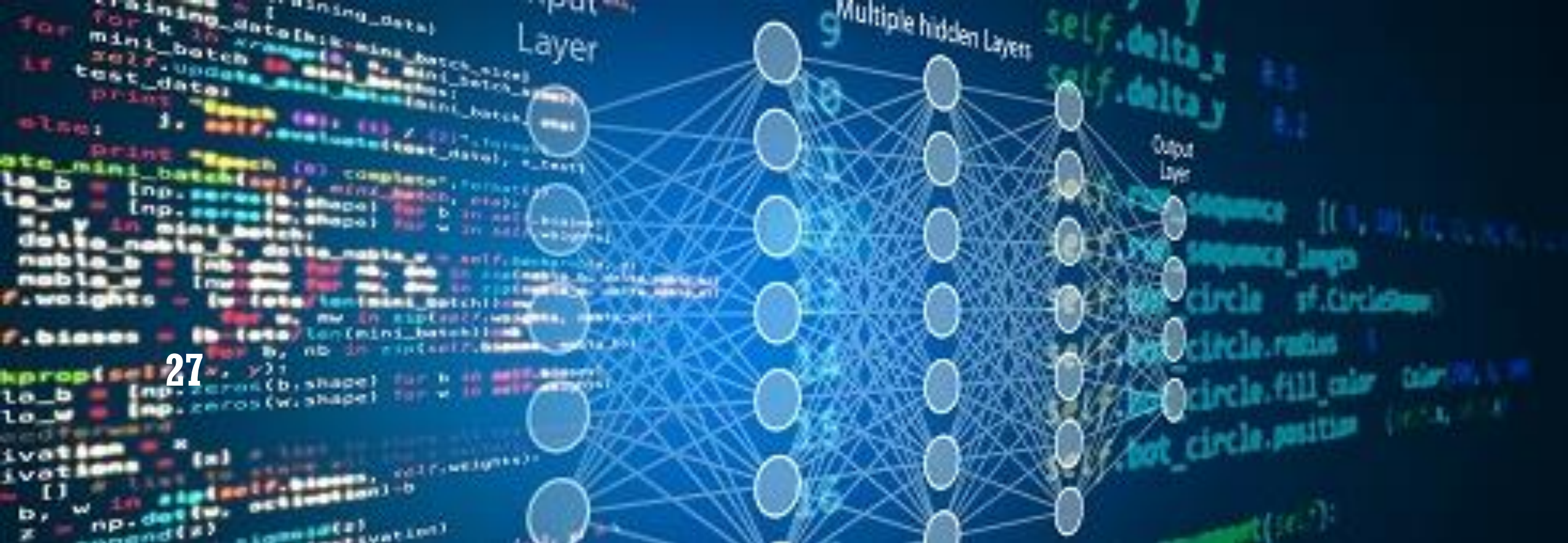
Simple Linear Regression

$$\hat{y} = \theta_0 + \theta_1 x$$

Predictions on a **scatter plot**.



Not a great linear fit visually? We'll come back to this!



CONSTANT MODEL + MAE

THE MODELING PROCESS: USING A DIFFERENT LOSS FUNCTION

1. Choose a model



Constant Model

$$\hat{y} = \theta_0$$

2. Choose a loss function



L1 Loss

Mean Absolute Error (MAE)

Let's repeat, but with **Mean Absolute Error (MAE)** as our average loss.

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

3. Fit the model

Minimize average loss with calculus

4. Evaluate model performance

THE MODELING PROCESS: USING A DIFFERENT LOSS FUNCTION



1. Choose a model

Constant Model

$$\hat{y} = \theta_0$$



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3. Fit the model

Minimize
average loss
with calculus

How does this step change?

4. Evaluate model performance

FIT THE MODEL: REWRITE MAE FOR THE CONSTANT MODEL

Recall that Mean **Absolute** Error (MAE) is average **absolute** loss (L1 loss) over the data $\mathcal{D} = \{y_1, y_2, \dots, y_n\}$:

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n \underbrace{|y_i - \hat{y}_i|}_{\text{L1 loss on a single datapoint}}$$

Given the **constant model** $\hat{y} = \theta_0$:

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0|$$

We **fit the model** by finding the optimal $\hat{\theta}_0$ that minimizes the MAE.

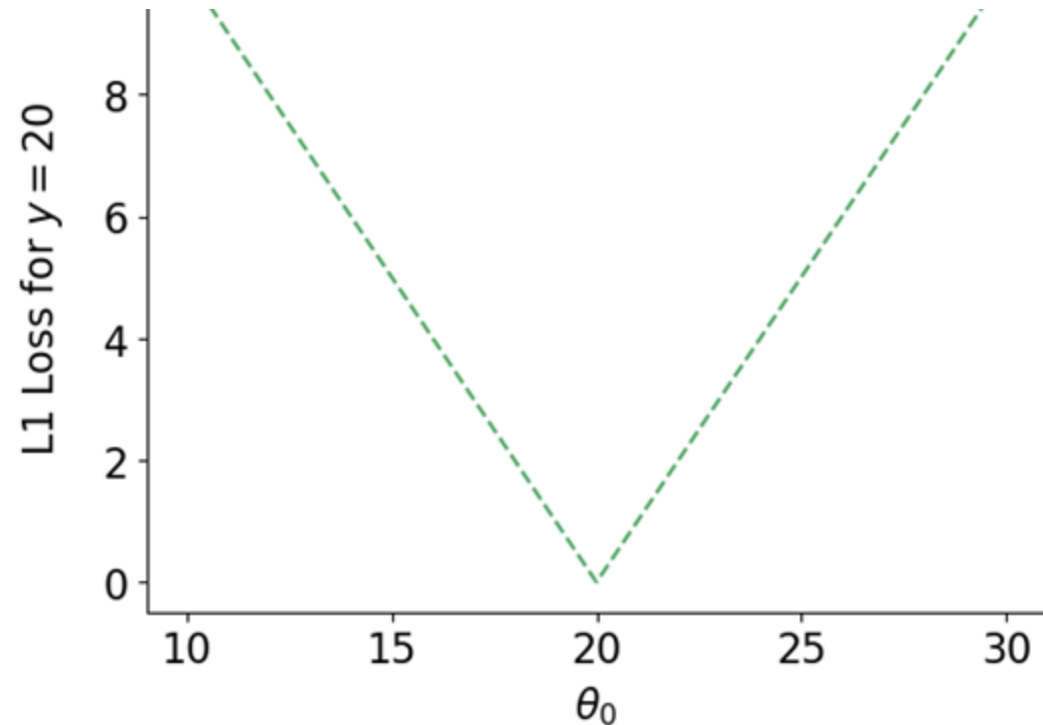
EXPLORING MAE: A PIECEWISE FUNCTION

Boba dataset {20, 21, 22, 29, 33} (Mean=25)

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0|$$

Absolute (L1) Loss on one observation:

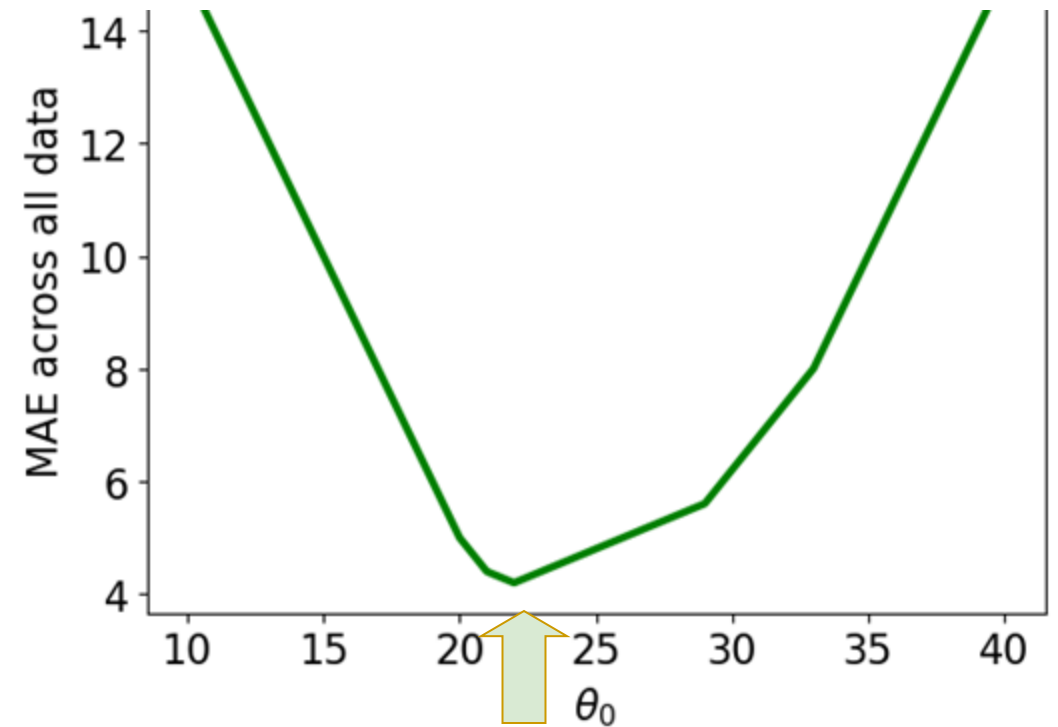
$$L_1(20, \theta_0) = |20 - \theta_0|$$



An absolute value curve,
centered at $\hat{\theta}_0 = 20$.

MAE (Mean Absolute Error) across all data:

$$\hat{R}(\theta_0) = \frac{1}{5} (|20 - \theta_0| + |21 - \theta_0| + |22 - \theta_0| + |29 - \theta_0| + |33 - \theta_0|)$$



Piecewise linear function...
minimized at... $\hat{\theta}_0 = 22$?

FIT THE MODEL: CALCULUS

1. Differentiate with respect to $\hat{\theta}_0$.

$$\frac{d}{d\theta_0} R(\theta_0) = \frac{d}{d\theta_0} \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0|$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{d}{d\theta_0} |y_i - \theta_0|$$

$$|y_i - \theta_0| = \begin{cases} y_i - \theta_0 & \text{if } \theta_0 \leq y_i \\ \theta_0 - y_i & \text{if } \theta_0 > y_i \end{cases}$$

$$\frac{d}{d\theta_0} |y_i - \theta_0| = \begin{cases} -1 & \text{if } \theta_0 < y_i \\ 1 & \text{if } \theta_0 > y_i \end{cases}$$

$$= \frac{1}{n} \left[\sum_{\theta_0 < y_i} (-1) + \sum_{\theta_0 > y_i} (+1) \right]$$

2. Set equal to 0.

$$0 = \frac{1}{n} \sum_{\hat{\theta}_0 < y_i} (-1) + \frac{1}{n} \sum_{\hat{\theta}_0 > y_i} (1)$$

3. Solve for $\hat{\theta}_0$.

$$0 = -\frac{1}{n} \sum_{\hat{\theta}_0 < y_i} (1) + \frac{1}{n} \sum_{\hat{\theta}_0 > y_i} (1)$$

$$\sum_{\hat{\theta}_0 < y_i} (1) = \sum_{\hat{\theta}_0 > y_i} (1)$$

MEDIAN MINIMIZES MAE FOR THE CONSTANT MODEL

The constant model parameter $\theta = \theta_0$ that minimizes MAE must satisfy:

$$\underbrace{\sum_{\hat{\theta}_0 < y_i} (1)}_{\substack{\text{\# observations} \\ \text{greater than } \hat{\theta}_0}} = \underbrace{\sum_{\hat{\theta}_0 > y_i} (1)}_{\substack{\text{\# observations} \\ \text{less than } \hat{\theta}_0}}$$

$\hat{\theta}_0$ needs to be located where there are **an equal number of y_i 's to the left and right**.

This is the definition of the **median**!

$$\hat{\theta}_0 = \text{median}(y)$$

For example, in our boba tea dataset $\{20, 21, 22, 29, 33\}$, the value **22** is the median.



THE MODELING PROCESS: USING A DIFFERENT LOSS FUNCTION

1. Choose a model



Constant Model

$$\hat{y} = \theta_0$$

2. Choose a loss function



L1 Loss

Mean Absolute Error
(MAE)

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0|$$

3. Fit the model



Minimize
average loss
with calculus

$$\hat{\theta}_0 = \text{median}(y)$$

**4. Evaluate model
performance**

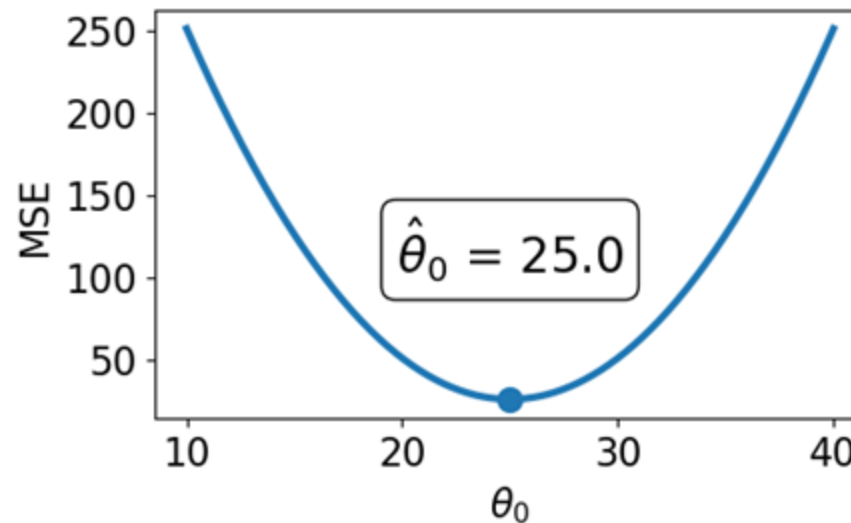
MSE AND MAE: COMPARING OPTIMAL PARAMETERS

MSE (Mean Squared Error)

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0)^2$$

Minimized with **sample mean**:

$$\hat{\theta}_0 = \text{mean}(y) = \bar{y}$$

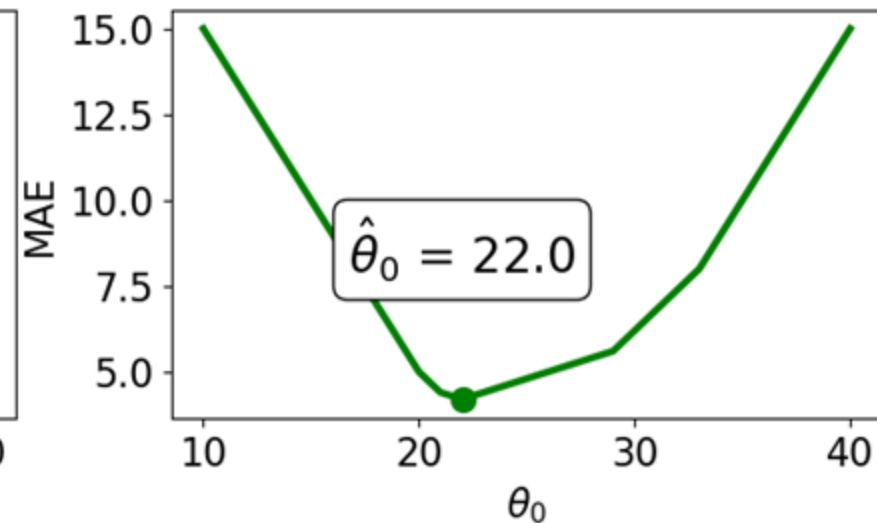


MAE (Mean Absolute Error)

$$\hat{R}(\theta_0) = \frac{1}{n} \sum_{i=1}^n |y_i - \theta_0|$$

Minimized with **sample median**:

$$\hat{\theta}_0 = \text{median}(y)$$



MSE AND MAE: FINDING MINIMIZING VALUES

MSE (Mean Squared Error)

Minimized with **sample mean**:

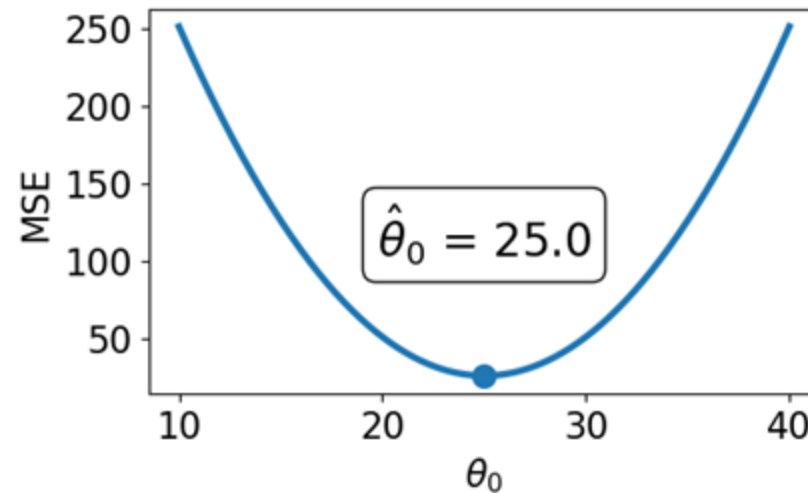
$$\hat{\theta}_0 = \text{mean}(y) = \bar{y}$$

MAE (Mean Absolute Error)

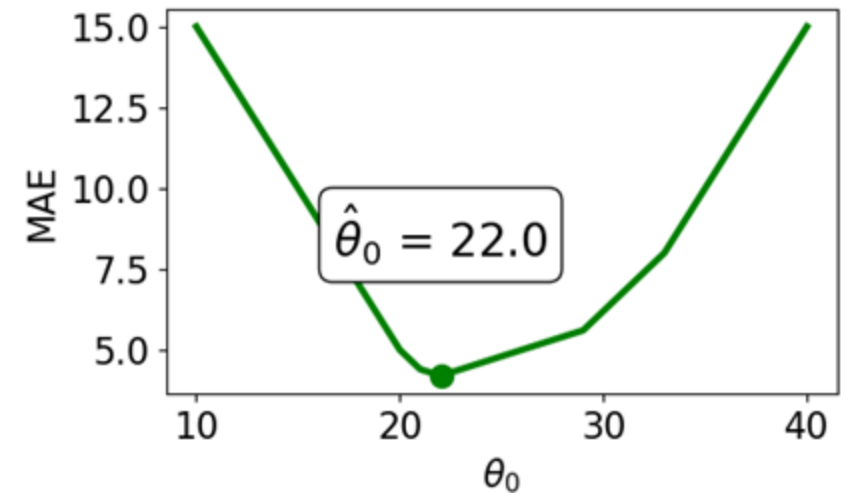
Minimized with **sample median**:

$$\hat{\theta}_0 = \text{median}(y)$$

data = {20, 21, 22, 29, 33}



Smooth. Easy to minimize using numerical methods.



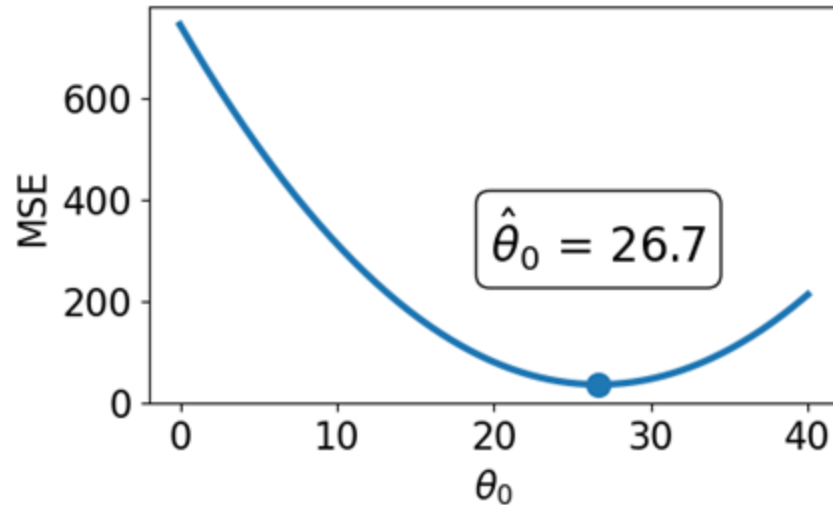
⚠ Piecewise. at each of the “kinks,” it’s not differentiable. Harder to minimize.

MSE AND MAE: COMPARING UNIQUENESS OF SOLUTIONS

MSE (Mean Squared Error)

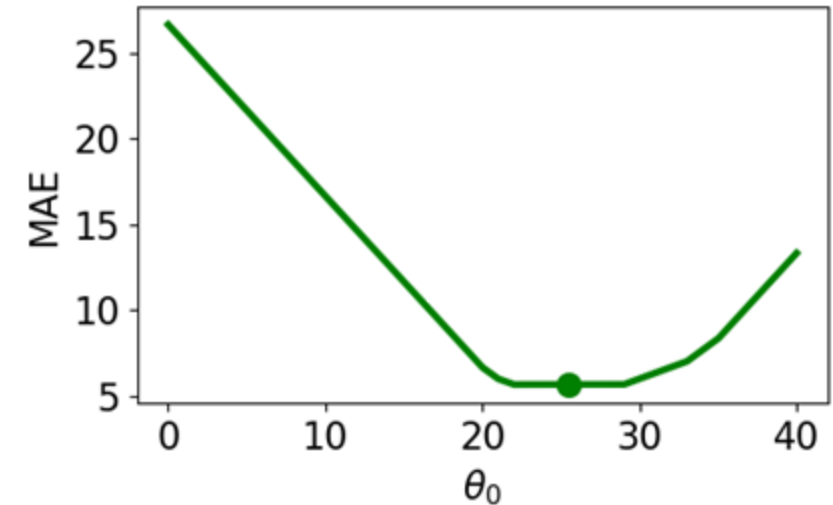
MAE (Mean Absolute Error)

Suppose we add a 6th observation to our bubble tea dataset:
 $\{20, 21, 22, 29, 33, \mathbf{35}\}$



Unique $\hat{\theta}_0$:

$$\hat{\theta}_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i \right)$$



⚠ Infinitely many $\hat{\theta}_0$ s. Any $\hat{\theta}_0$ in range (22, 29) minimizes MAE. Equal # of y_i 's to left and right.

We typically choose the average of the boundary values (25.5).

MSE AND MAE: COMPARING SENSITIVITY TO OUTLIERS

MSE (Mean Squared Error)

Minimized with **sample mean**:

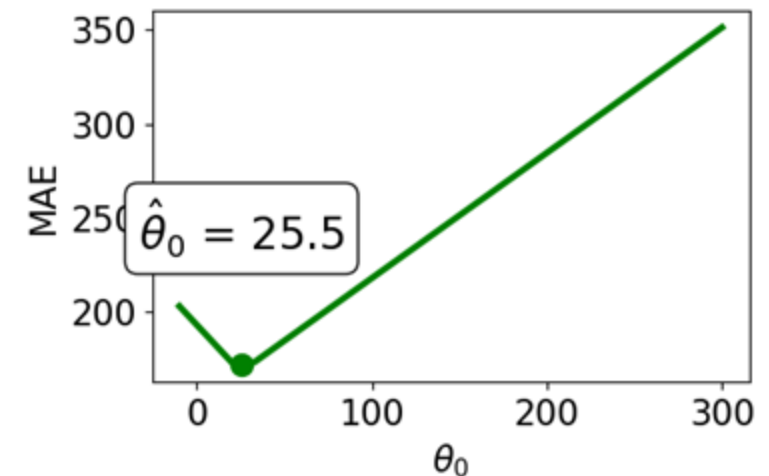
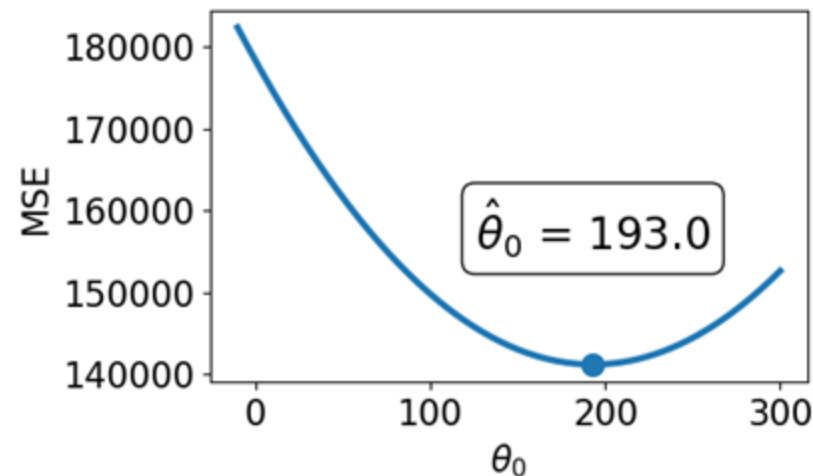
$$\hat{\theta}_0 = \text{mean}(y) = \bar{y}$$

MAE (Mean Absolute Error)

Minimized with **sample median**:

$$\hat{\theta}_0 = \text{median}(y)$$

data = {20, 21, 22, 29, 33, **1033**}



⚠ Sensitive to outliers (since they change mean substantially). Sensitivity also depends on the dataset size.

More robust to outliers.



TRANSFORMATIONS OF LINEAR MODELS

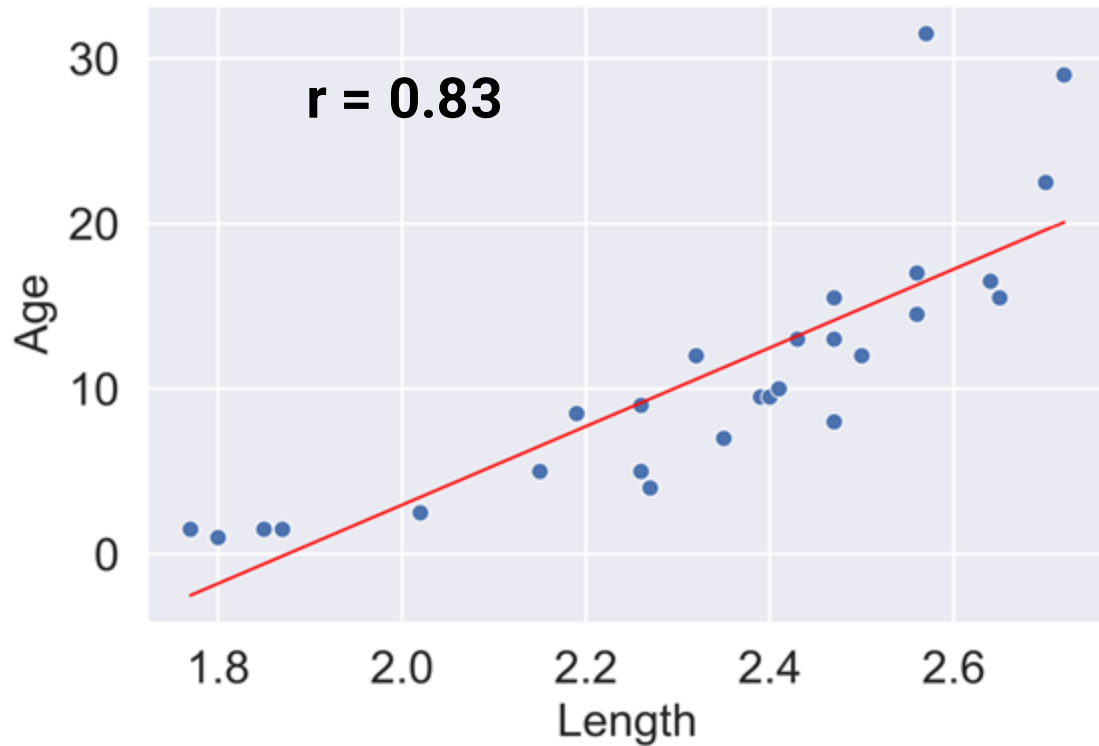
BACK TO LEAST SQUARES REGRESSION WITH DUGONGS



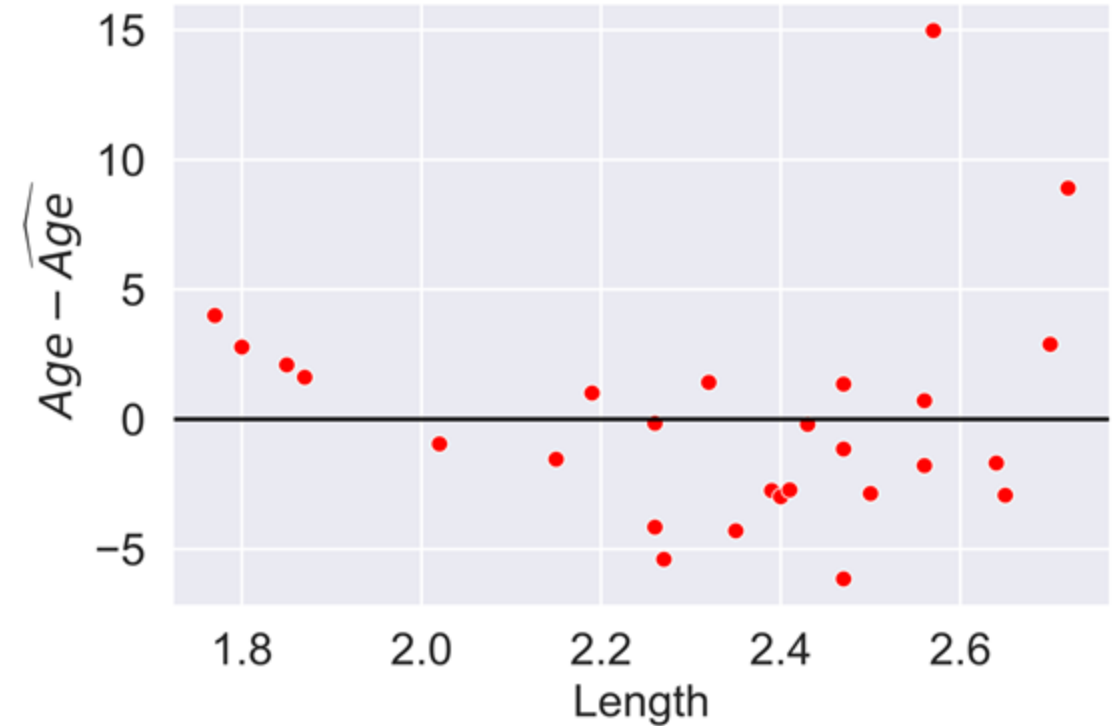
The residual plot of a good regression shows **no pattern**.

BACK TO LEAST SQUARES REGRESSION WITH DUGONGS

Age by Length



Residual Plot



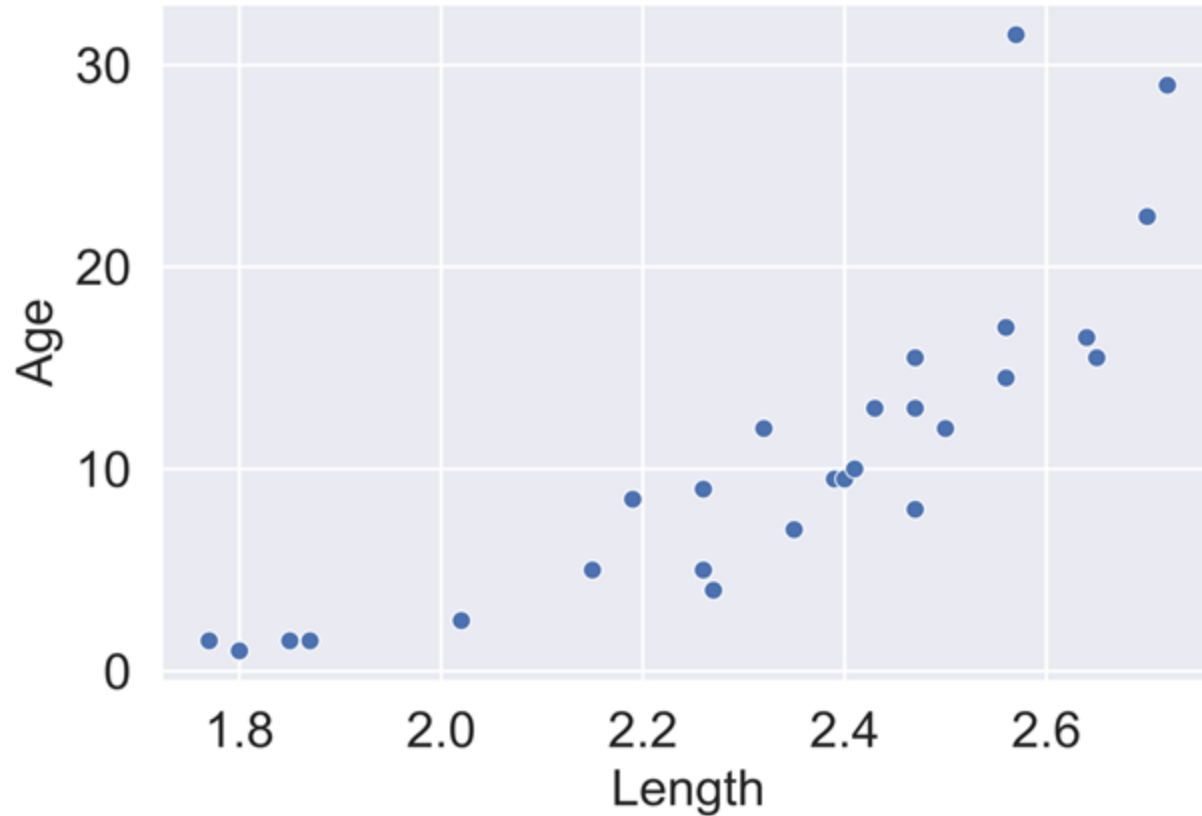
The scatter plot is **non-linear**, so the SLR **residual plot** shows a pattern indicative of a bad fit.

To improve the fit, we can **linearize** the data with a transformation.

POSSIBLE TRANSFORMATIONS

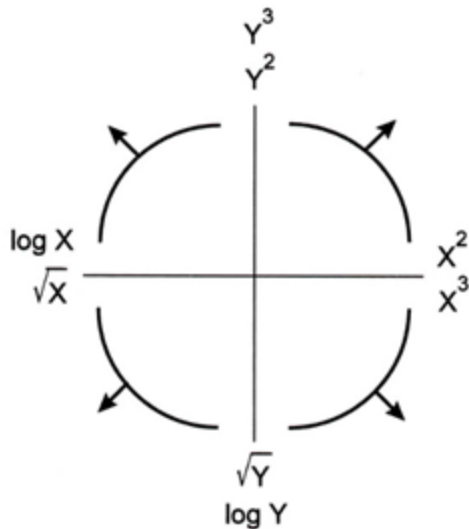
Large age values appear **unconstrained**.

Squish differences with **log(Age)** transform.



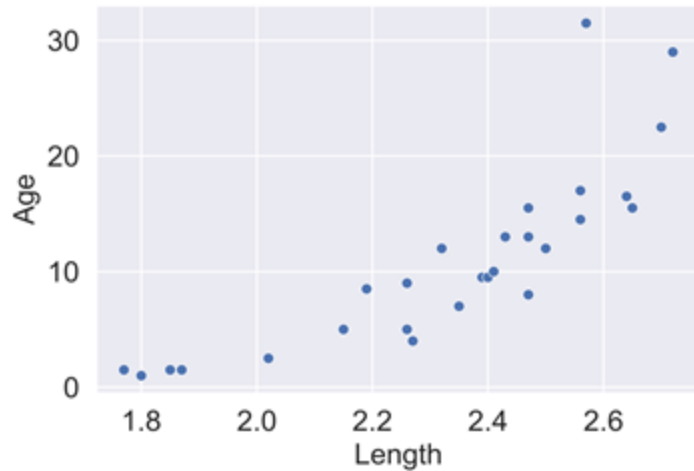
Large length values appear **compressed**.

Magnify differences with **(Length)²** transform.

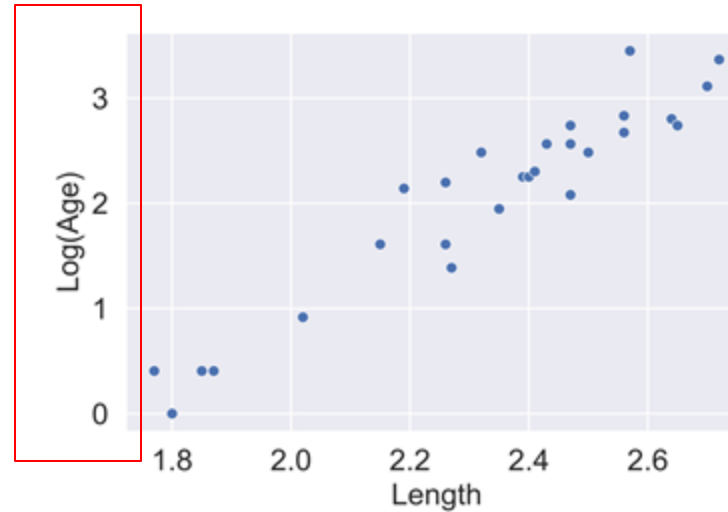


EFFECT OF LOG AND SQUARING TRANSFORMATIONS

No transformation
[Raw; natural scale]

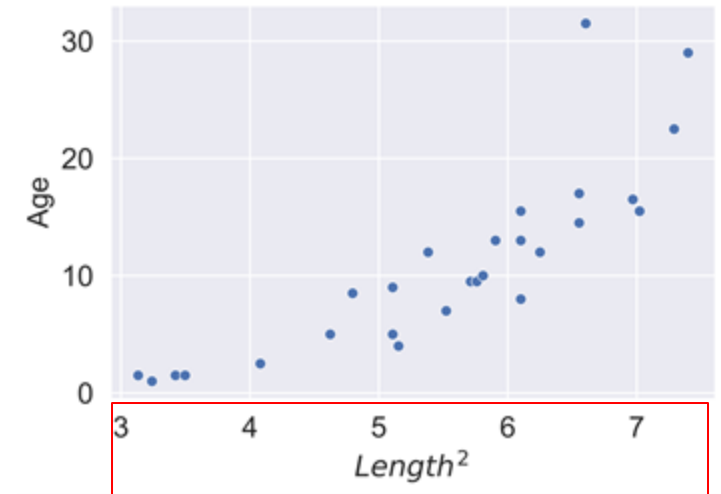


Log transformation



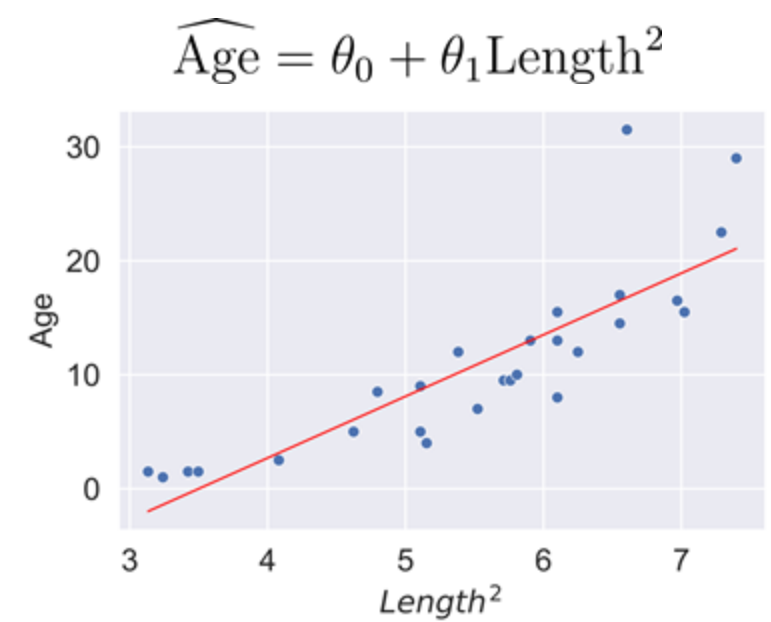
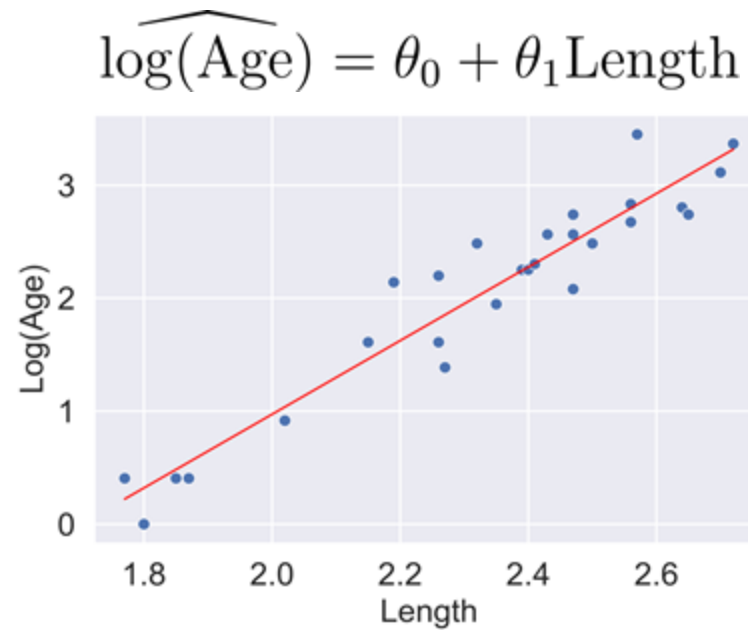
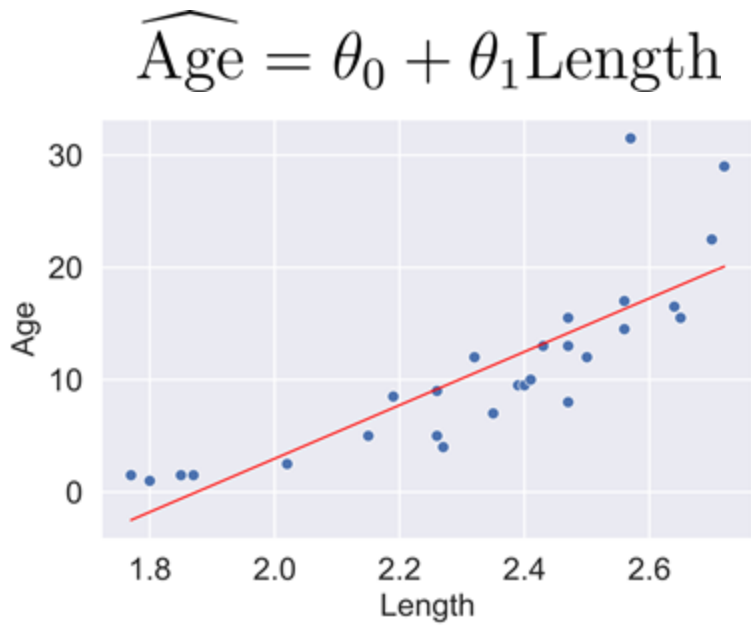
Big improvement in linearity.

Squaring transformation



Very subtle improvement in linearity.
Hard to see here—wait a few slides!

SLR OUTCOMES ALWAYS LINEAR IN THE PROVIDED FEATURE



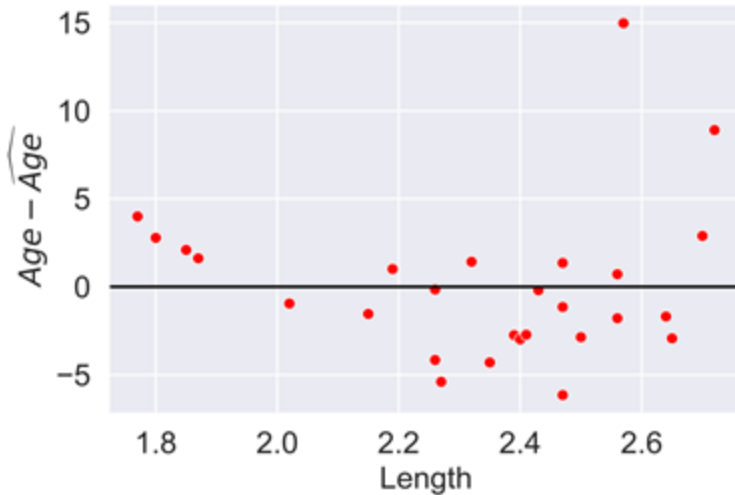
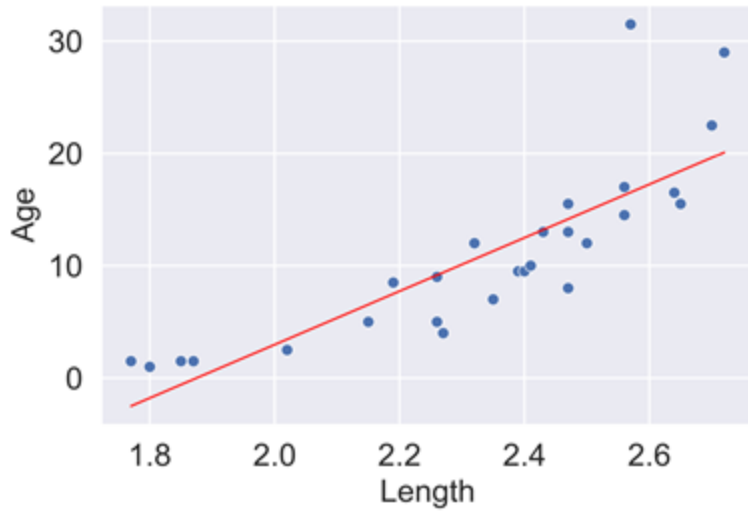
SLR doesn't "know" that the output is logged Age or the input is squared Length.

- To SLR, $\log(\text{Age})$ and Length^2 are just vectors of values!
- The outcome provided to SLR is always modeled as linear in the provided feature.

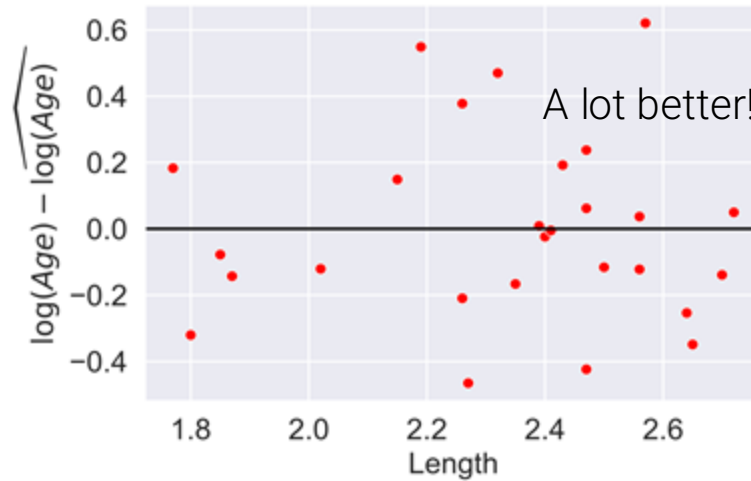
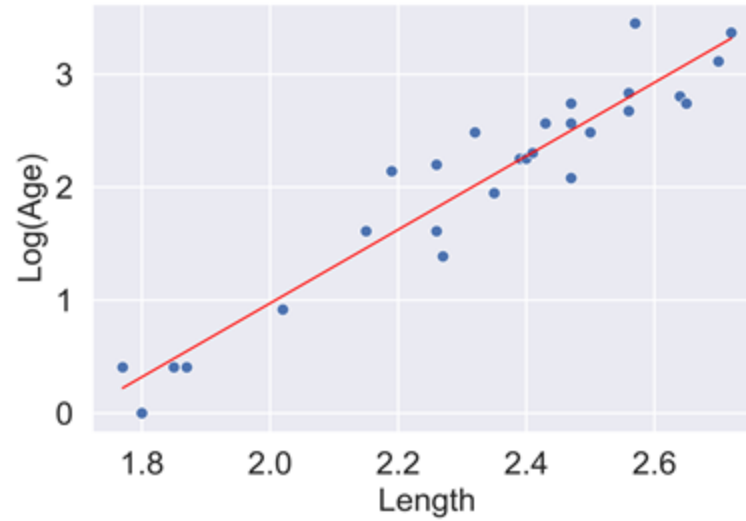
Note: When squaring a feature, we usually include the linear feature for extra flexibility (e.g., $\hat{Y} = \theta_0 + \theta_1 X + \theta_2 X^2$). Next lecture!

SLR OUTCOMES ALWAYS LINEAR IN THE PROVIDED FEATURE

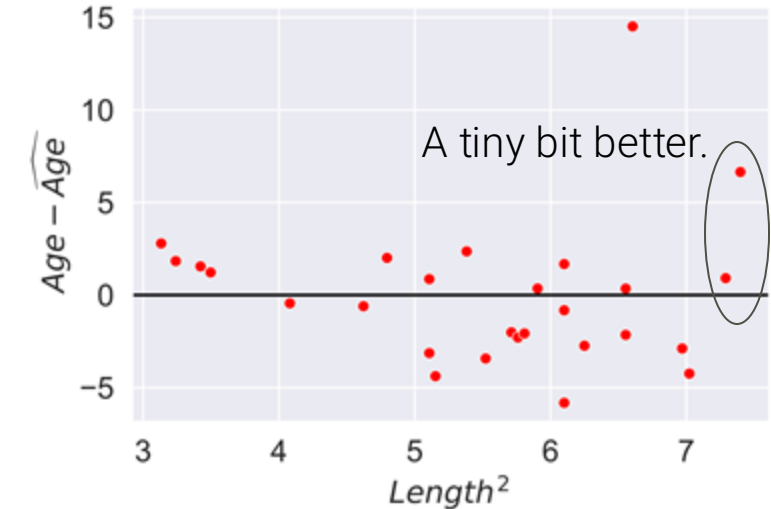
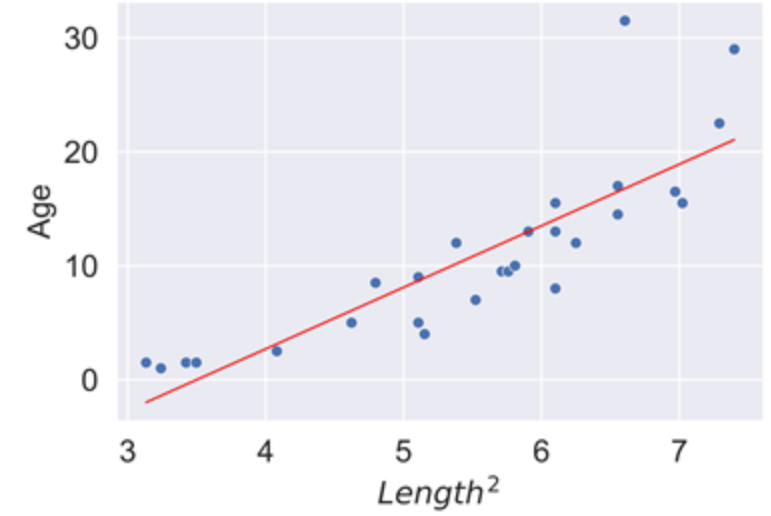
$$\widehat{\text{Age}} = \theta_0 + \theta_1 \text{Length}$$



$$\log(\widehat{\text{Age}}) = \theta_0 + \theta_1 \text{Length}$$



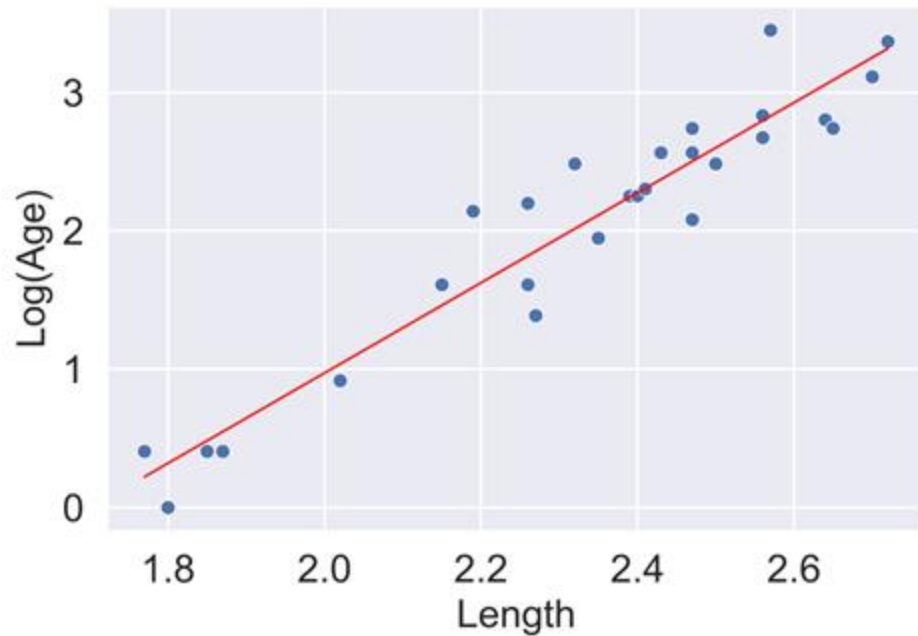
$$\widehat{\text{Age}} = \theta_0 + \theta_1 \text{Length}^2$$



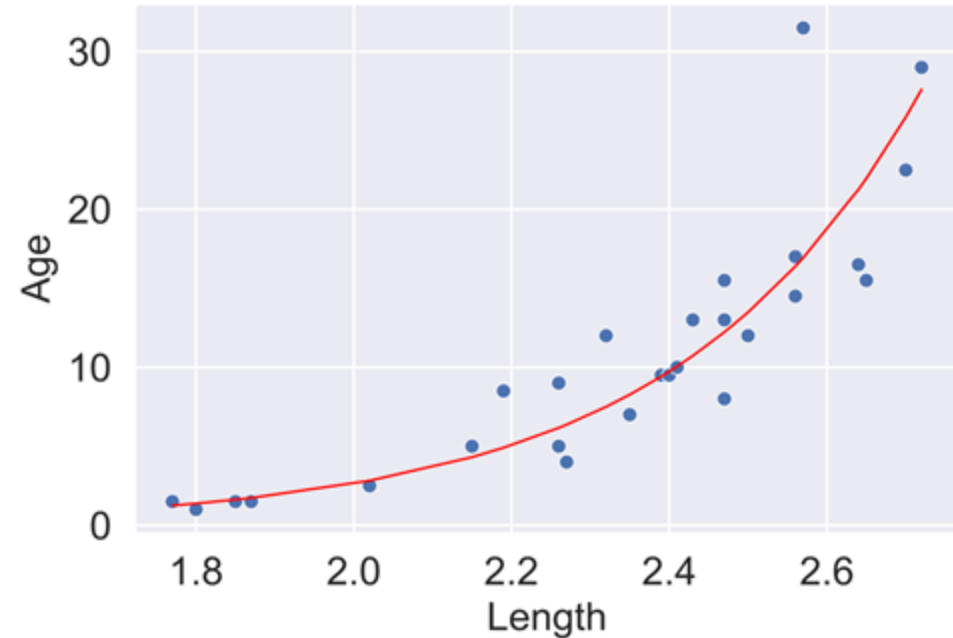
Note: When squaring a feature, we usually include the linear feature for extra flexibility (e.g., $\hat{Y} = \theta_0 + \theta_1 X + \theta_2 X^2$). Next lecture!

VISUALIZING LOG-TRANSFORMED PREDICTIONS ON NATURAL SCALE

$$\widehat{\log(\text{Age})} = \theta_0 + \theta_1 \text{Length}$$



$$\widehat{\text{Age}} = e^{\theta_0 + \theta_1 \text{Length}}$$

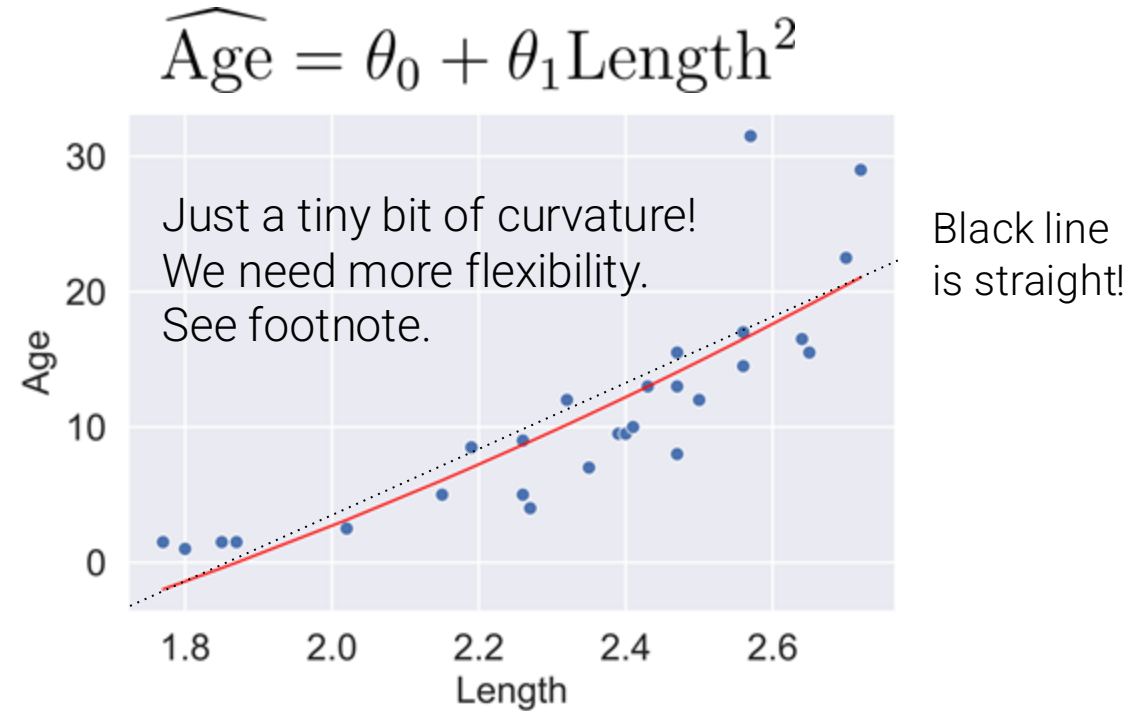
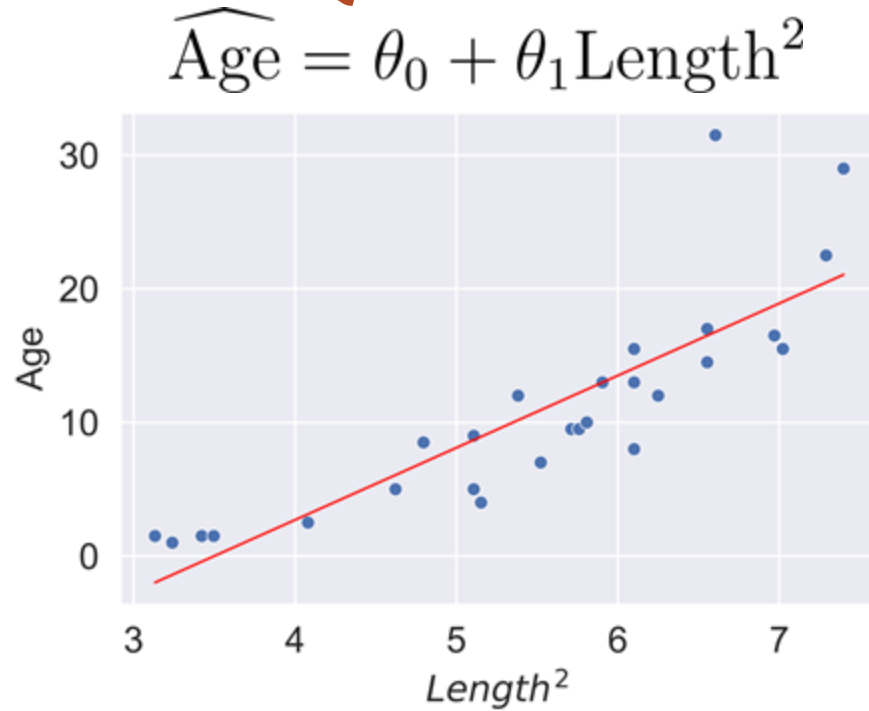


Unlike SLR, we know that the y-variable is $\log(\text{Age})$.

- To get Age predictions, we exponentiate the $\log(\text{Age})$ predictions.

$\log(\text{Age})$ predictions are linear in Length. Age predictions are not linear in Length.

VISUALIZING SQUARED FEATURES ON NATURAL SCALE



Unlike SLR, we know that the x-variable is Length^2 !

- Predictions are already on the Age scale, so no need to change the y-axis.
- We can use the non-transformed feature (Length) on the x-axis to see non-linearities.

Age predictions are linear in Length^2 , but not linear in Length.

READING MATERIAL

- Chapter# 4: Principle of Data Science
- Chapter# 4: Hands-On ML [Aurélien]